Midterm Review

Computer Organization

Chandra Gummaluru

chandra-gummaluru.github.io



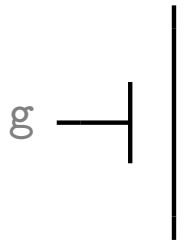
For each topic, we will **summarize** what you need to know, and then answer a few questions.

1. transistors
2. Boolean algebra
3. combinational design
4. logical devices
5. binary arithmetic
6. arithmetic circuits
7. flip-flops
8. state machines



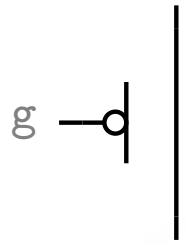
transistors

nMOS



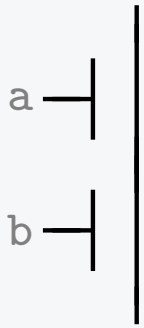
conducts when the gate is 1

pMOS



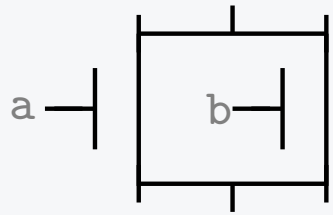
conducts when the gate is 0

in series



AND: both conduct

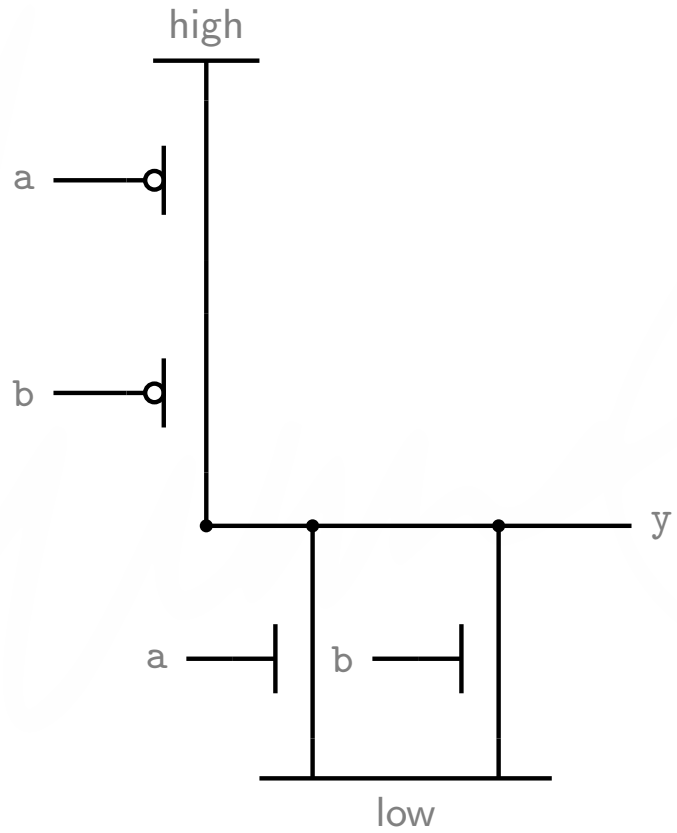
in parallel



OR: either conducts

a gate connects its output to high through one network and to low through the other

question 1

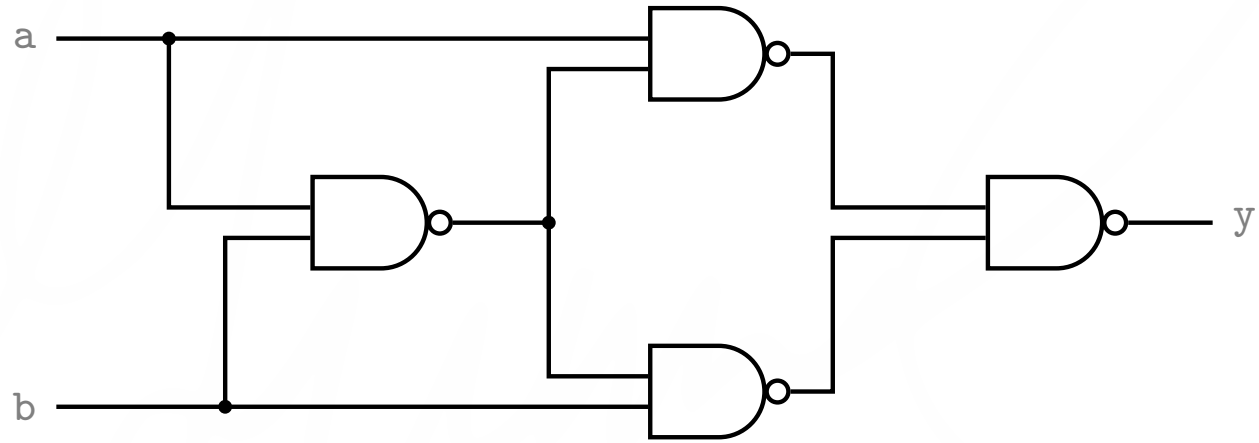


22 responses

Which gate does this transistor circuit implement?

A	AND	5%	1 vote
B	OR	5%	1 vote
C	NAND	59%	13 votes
✓	NOR	32%	7 votes

question 2



$$([a'+b][b'+a])' = [a'+b]' + [b'+a]'$$
$$= ab' + ba'$$

17 responses

Which gate do these four `NAND` gates build?

A	AND	12%	2 votes
B	OR	6%	1 vote
✓	XOR	53%	9 votes
D	XNOR	29%	5 votes

2

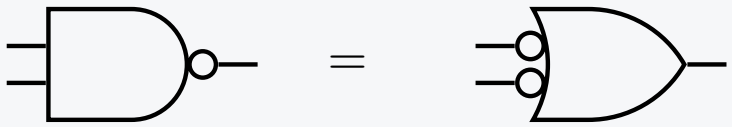
Boolean algebra

Each law comes in an AND form and an OR form.

identity	$a \cdot 1 = a$	$a + 0 = a$
null	$a \cdot 0 = 0$	$a + 1 = 1$
idempotent	$a \cdot a = a$	$a + a = a$
complement	$a \cdot \overline{a} = 0$	$a + \overline{a} = 1$
involution	$\overline{\overline{a}} = a$	
commutative	$ab = ba$	$a + b = b + a$
associative	$a(bc) = (ab)c$	$a + (b + c) = (a + b) + c$
distributive	$a(b + c) = ab + ac$	$a + bc = (a + b)(a + c)$
absorption	$a(a + b) = a$	$a + ab = a$
consensus	$ab + \overline{a}c + bc = ab + \overline{a}c$	
De Morgan	$\overline{a \cdot b} = \overline{a} + \overline{b}$	$\overline{a + b} = \overline{a} \cdot \overline{b}$

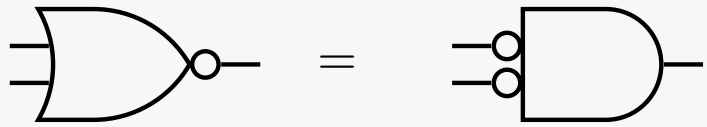
De Morgan's laws let us push a bubble through a gate: `AND` and `OR` swap, and every input and output flips.

first form



$$\overline{a \cdot b} = \overline{a} + \overline{b}$$

second form

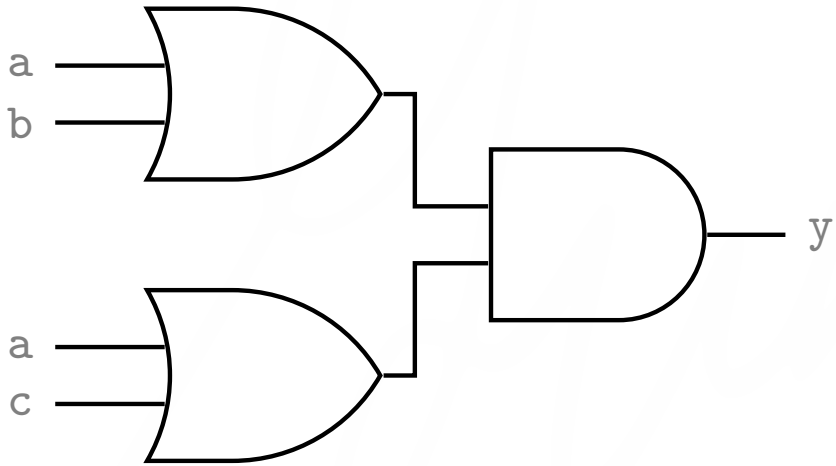


$$\overline{a + b} = \overline{a} \cdot \overline{b}$$

a `NAND` is an `OR` of inverted inputs; a `NOR` is an `AND` of inverted inputs

question 3

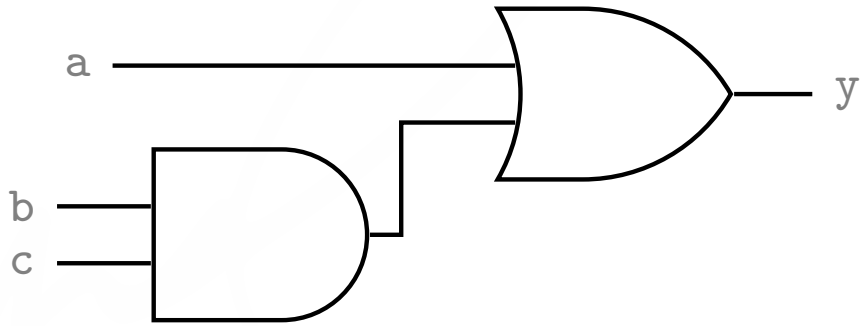
circuit A



(a+b)*(a+c)

a+(bc) = (bc)+a

circuit B



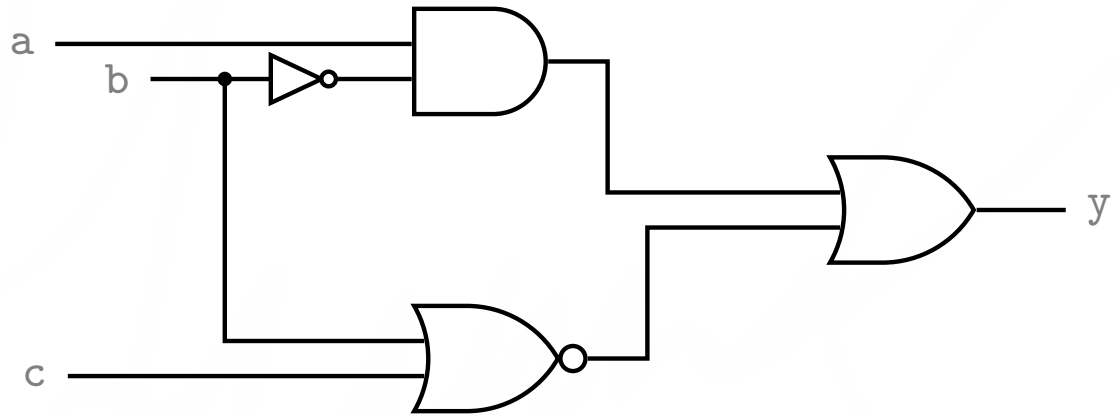
(b*c)+a

17 responses

Circuits A and B compute the same function.

✓ True	94% 16 votes	B False	6% 1 vote

question 4



(ab')+(b'c')

b'(a+c')

(ab')+(b+c)'

(ab')+(b+c)'

20 responses

Which expression does this circuit implement?

A $a\overline{b} + \overline{b} + \overline{c}$ **15%** 3 votes

B $a\overline{b} + \overline{b}, \overline{c}$ **70%** 14 votes

C $\overline{ab} + bc$ **10%** 2 votes

D $(a + \overline{b}), \overline{b}, \overline{c}$ **5%** 1 vote

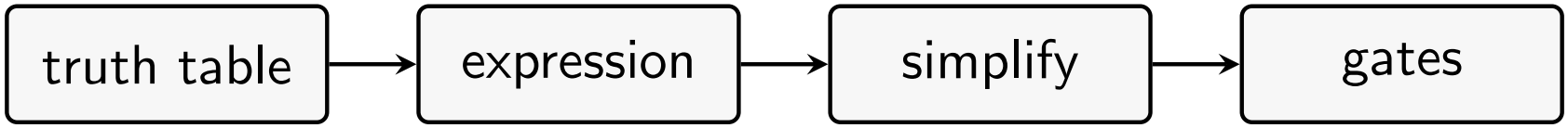
18 responses

Simplify $\overline{(\overline{a}+b),\overline{c}}+\overline{a}c$.

A	$a+c$		0% 0 votes
B	$(\overline{a}+b),\overline{c}$	(a'+b)'+c+a'c ab'+c+a'c ab'+c	0% 0 votes
✓	$a\overline{b}+c$		83% 15 votes
D	$a\overline{b}+\overline{a}c$		17% 3 votes

3

combinational design



a	b	c	f	
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	0	M_2
0	1	1	0	M_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	M_6
1	1	1	1	m_7

minterm m_i (the shaded rows)

AND of every input: $m_5 = \mathtt{a}\,\overline{\mathtt{b}}\,\mathtt{c}$

want these rows

$$\mathtt{f} = \Sigma m(0, 1, 4, 5, 7)$$

maxterm M_i (the white rows)

OR of every input: $M_2 = \mathtt{a} + \overline{\mathtt{b}} + \mathtt{c}$

don't want these rows

$$\mathtt{f} = \Pi M(2, 3, 6)$$

$M_i = \overline{m_i}$, so the two lists of the same function name opposite rows

sum of products

group the 1s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = \bar{b} + ac$$

product of sums

group the 0s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = (a + \bar{b})(\bar{b} + c)$$

size $1, 2, 4, 8, \ldots$

1	1	0	0
1	1	0	0

as large and as few as possible

wrap around

1	0	0	1
1	0	0	1

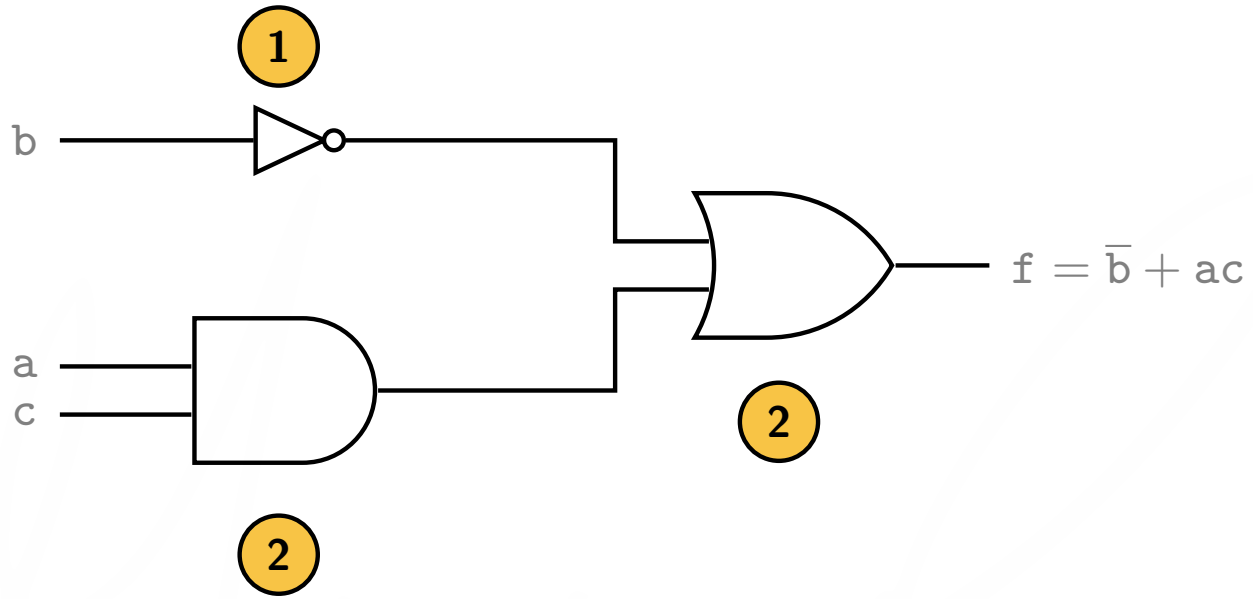
the edges are neighbours

don't-cares

0	1	X	0
0	1	1	0

an X joins a group if it helps

The **cost** of a circuit can be counted in three ways.



the circles count the inputs of each gate

L = 3

literals in the expression

$\bar{b}$, a, c

G = 4

gate inputs, not counting NOTs

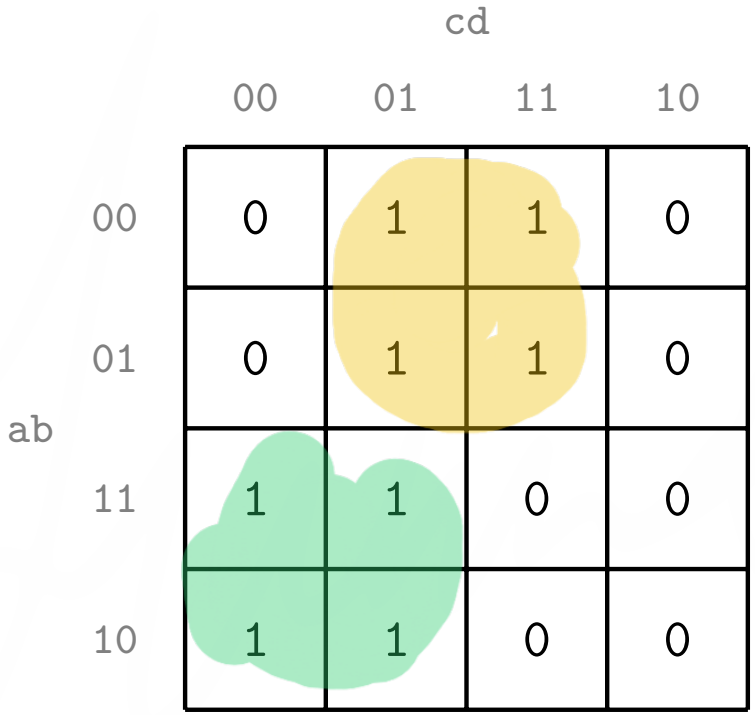
$2 + 2$

GN = 5

gate inputs, NOTs included

$2 + 2 + 1$

question 6



ac'+a'd

find the minimal sum of products

18 responses

What is the minimal sum of products for this Karnaugh map?

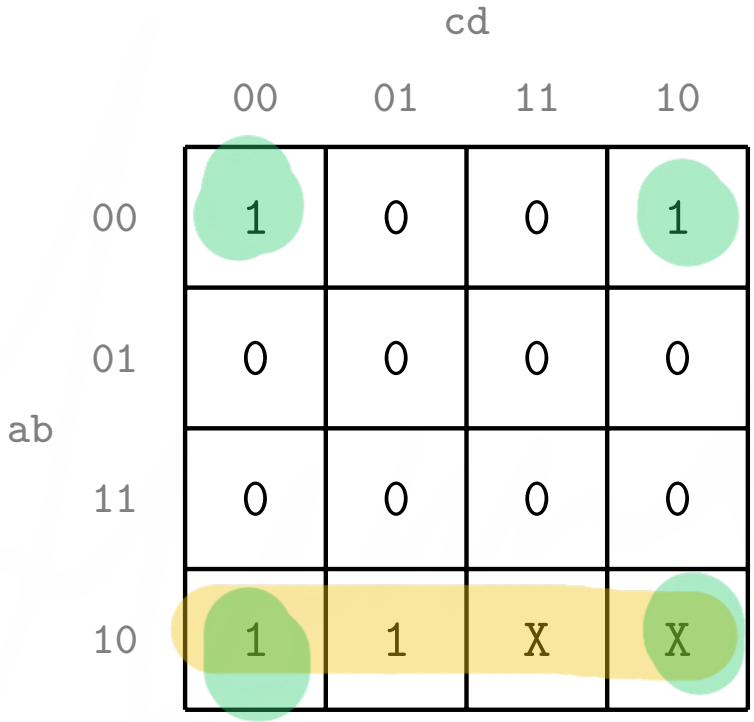
A $\bar{a}d + \bar{c}d$ — 0% (0 votes)

✓ $\bar{a}d + a\bar{c}$ — 83% (15 votes)

C $d + a\bar{c}$ — 17% (3 votes)

D $\bar{a}d + ab\bar{c}$ — 0% (0 votes)

question 7



ab' + b'd'

find the minimal sum of products

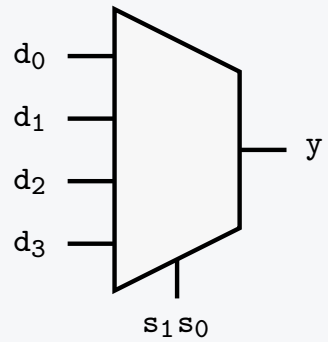
Audience Response

Interactive widget

4

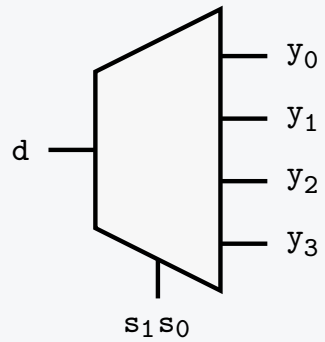
logical devices

multiplexer



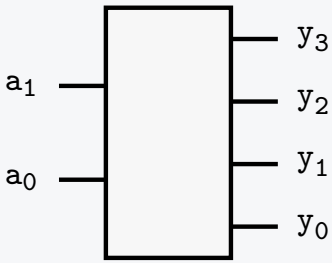
k select bits pick one of 2^k inputs

demultiplexer



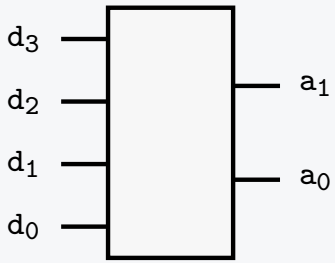
routes d to the selected output

decoder



n bits $\rightarrow$ one of 2^n outputs is 1; $y_i = m_i$

encoder

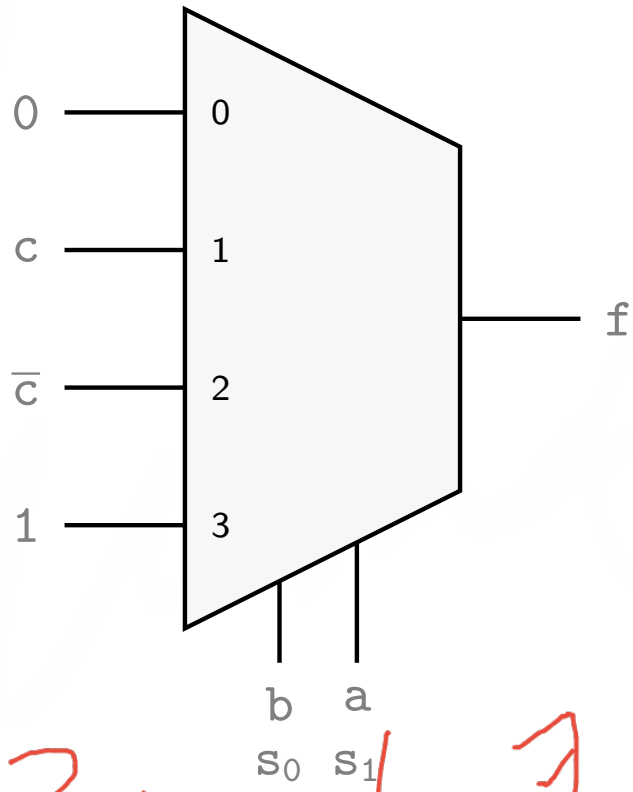


the number of the one input that is 1

Audience Response

Interactive widget

question 9



3, 4, 6, 7

a, b, c

a	b	c	
0	0	0	= 0
0	0	1	= 0
0	1	0	= 0
0	1	1	= 1
1	0	0	= 1
1	0	1	= 0
1	1	0	= 1
1	1	1	= 1

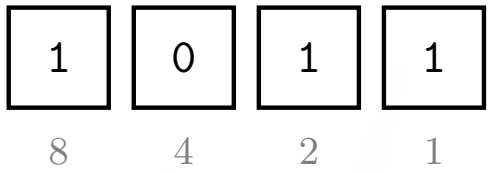
Audience Response

Interactive widget

5

binary arithmetic

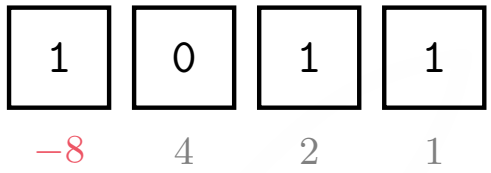
unsigned



$$8 + 2 + 1 = 11$$

range: 0 to $2^n - 1$

signed



$$-8 + 2 + 1 = -5$$

range: -2^{n-1} to $2^{n-1} - 1$

decimal to binary

$11 = 1011$

halve repeatedly; the remainders, read upward, are the bits

binary and hex

1011 0110 $\leftrightarrow$ B6

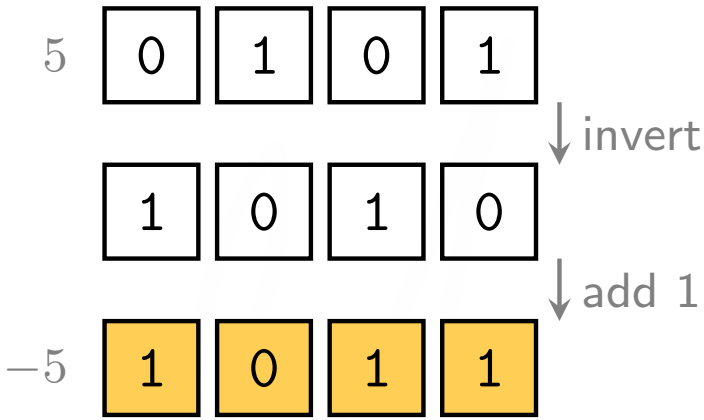
one hex symbol per group of 4 bits; A $= 10, \ldots,$ F $= 15$

binary to decimal

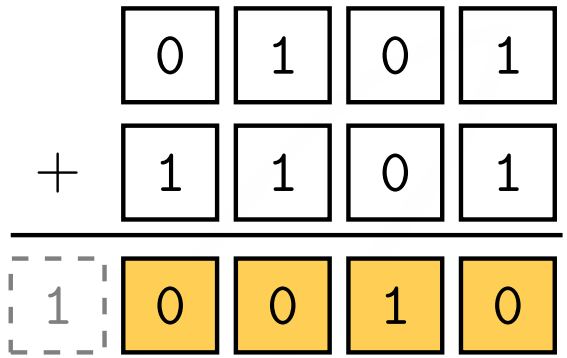
0110 $= 4 + 2 = 6$

add the weights of the 1s; signed, the top weight is negative

negating



adding



$5 + (-3) = 2$; the carry out is dropped

subtracting is adding the negation

$a - b = a + \overline{b} + 1$, the same bit operations for signed and unsigned numbers

00110
10101
11011

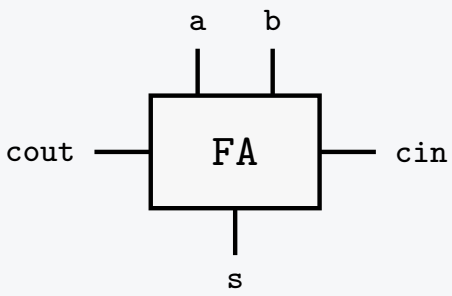
Audience Response

Interactive widget



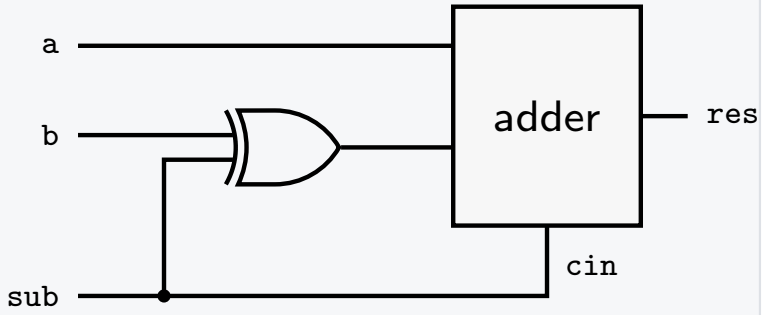
arithmetic circuits

full adder



$$\texttt{s} = \texttt{a} \oplus \texttt{b} \oplus \texttt{cin}$$
$$\texttt{cout} = \texttt{ab} + \texttt{cin}\,(\texttt{a} \oplus \texttt{b})$$

adder/subtractor



$$\texttt{sub} = 1\text{: } \texttt{a} + \overline{\texttt{b}} + 1 = \texttt{a} - \texttt{b}$$

unsigned: read `carry`

$$\begin{array}{r} 1100 \\ +\ 0110 \\ \hline 1\ 0010 \end{array}$$

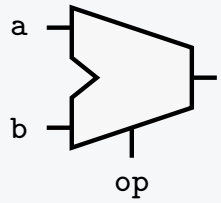
$12 + 6 \neq 2$: a + b is wrong when carry $= 1$
(a − b is wrong when carry $= 0$)

signed: read `overflow`

$$\begin{array}{r} 0110 \\ +\ 0011 \\ \hline 1001 \end{array}$$

$6 + 3 \neq -7$: two numbers of the same sign gave a result of the other sign

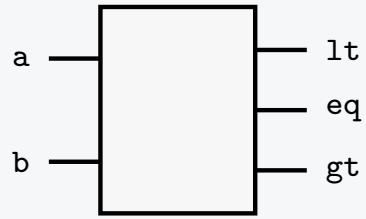
ALU



	useb	inv	cin
a	0	0	0
a + 1	0	0	1
a − 1	0	1	0
a + b	1	0	0
a − b	1	1	1

plus bitwise AND, OR, NOT;
a mux picks the result

comparator



1011 vs 1101: lt

the first bit from the left where they differ decides

shifter

1101 << 1 1010 $\times 2$

1101 >> 1 0110 logical

1101 >> 1 1110 arithmetic

multiplier

$$\begin{array}{r} 101 \\ \times\ 011 \\ \hline 101 \\ 1010 \\ \hline 1111 \end{array}$$

one shifted row per bit of b; the product needs $2n$ bits

signed: sign-extend, subtract the last row

Audience Response

Interactive widget

1100
1110
1010

0110
0011
1001

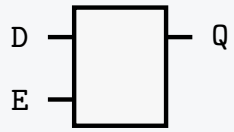
Audience Response

Interactive widget

7

latches and flip-flops

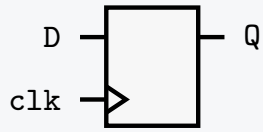
D latch



E	Q_t
0	Q_{t-1}
1	D

follows D while $E = 1$

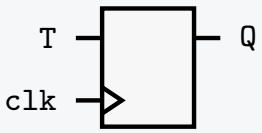
D flip-flop



D	Q_t
0	0
1	1

copies D on the clock edge

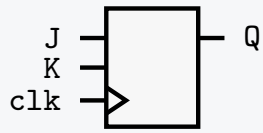
T flip-flop



T	Q_t
0	Q_{t-1}
1	$\overline{Q}_{t-1}$

toggles when $T = 1$: $D = T \oplus Q$

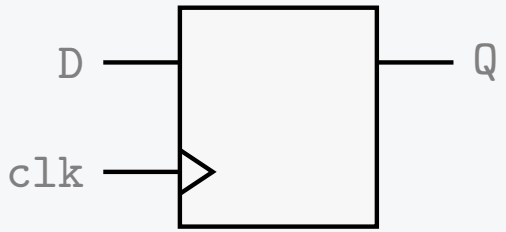
JK flip-flop



J	K	Q_t	
0	0	Q_{t-1}	hold
0	1	0	reset
1	0	1	set
1	1	$\overline{Q}_{t-1}$	toggle

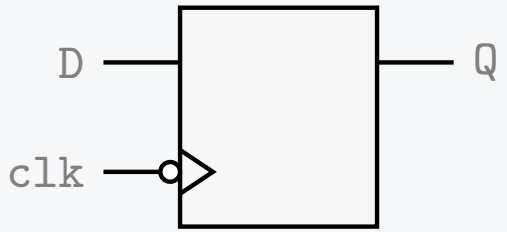
$D = J\overline{Q} + \overline{K}Q$

positive edge



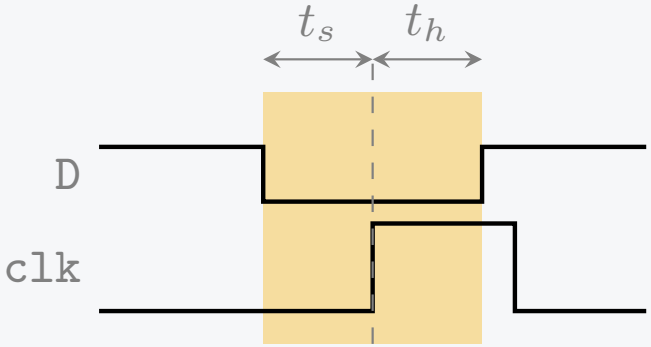
acts as clk rises

negative edge



acts as clk falls (the bubble)

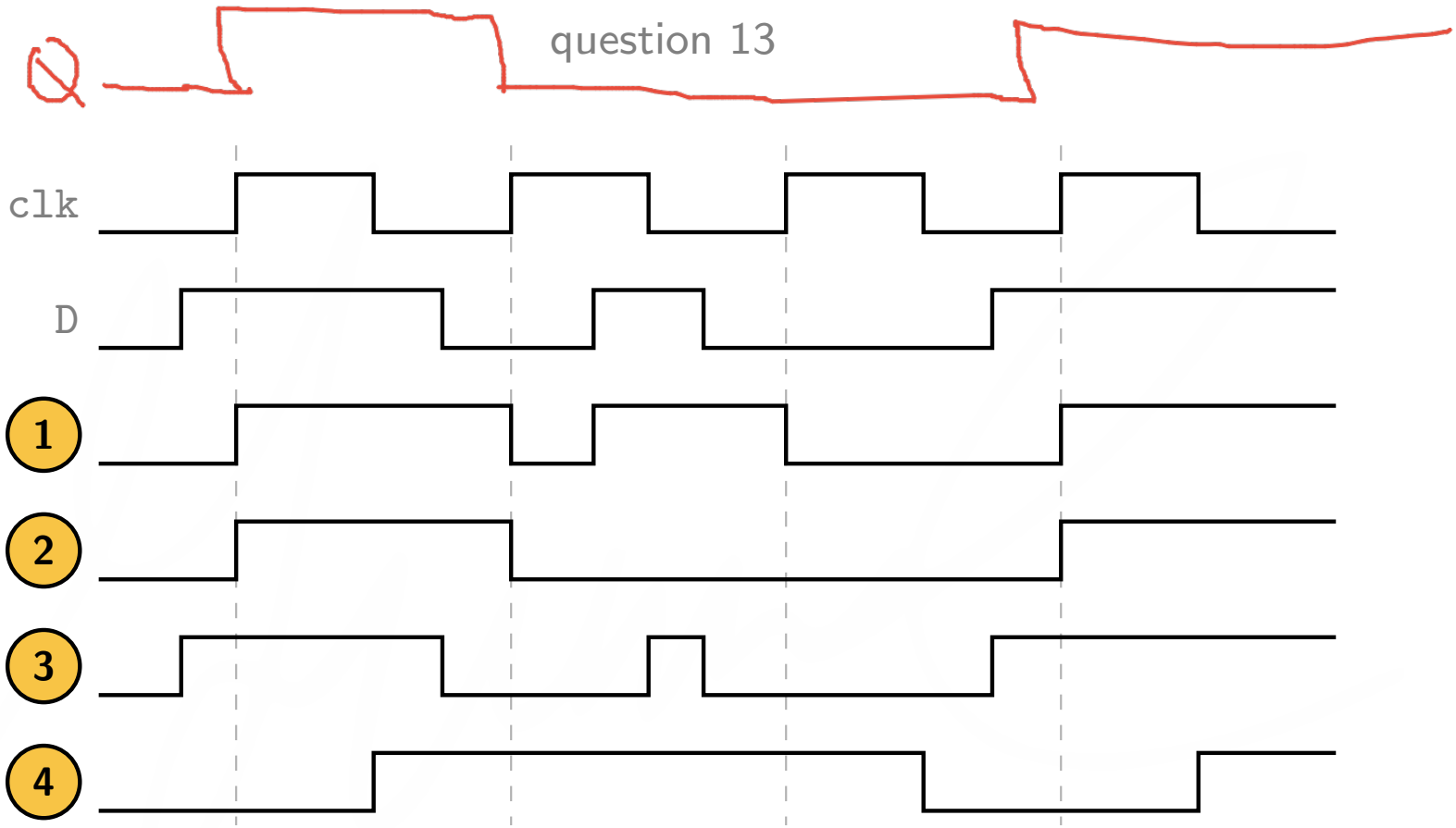
setup and hold times



setup time t_s: D is stable this long *before* the edge

hold time t_h: D stays stable this long *after* the edge

if D changes in the shaded window, the stored value is unreliable



Q starts at 0; which waveform is Q?

Audience Response

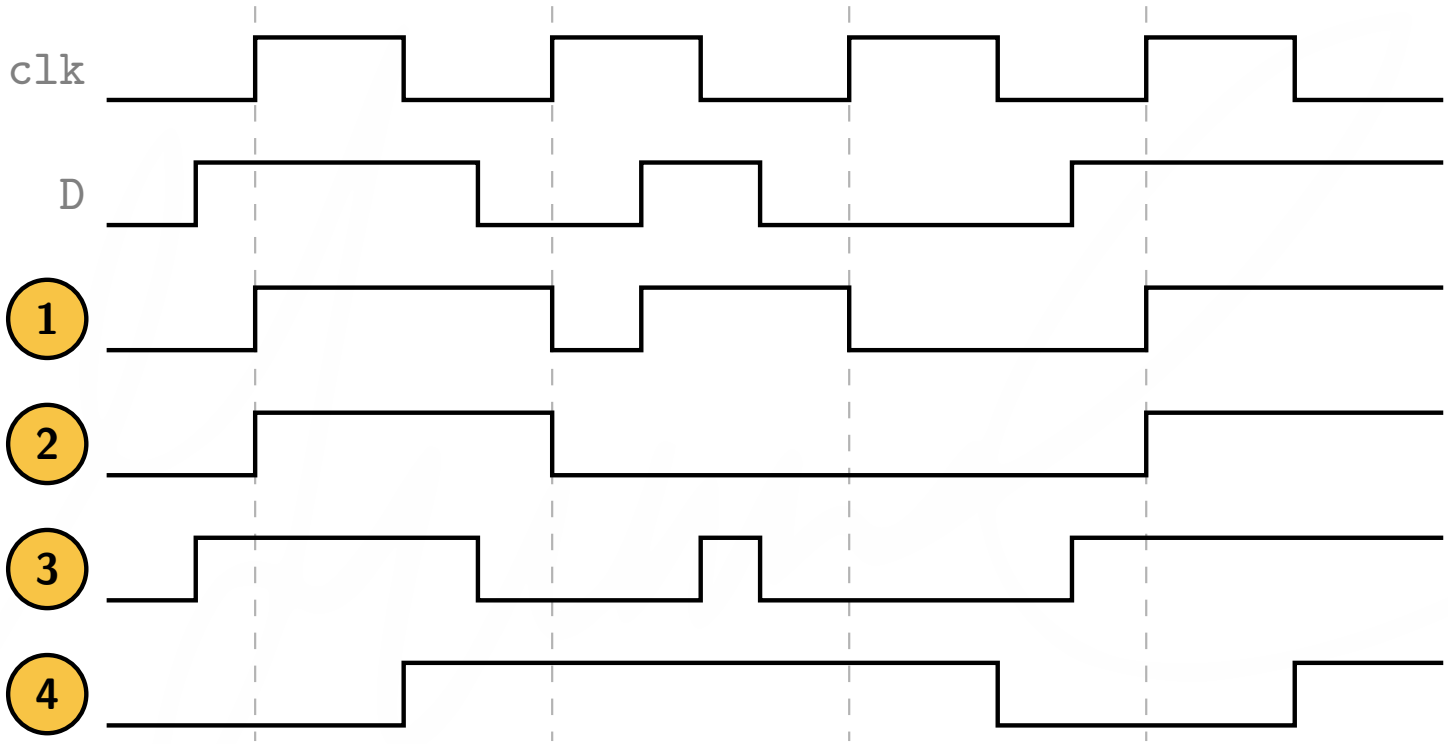
Interactive widget

Audience Response

Interactive widget

110010

question 15



`Q` starts at `0`; which waveform is `Q`?

Audience Response

Interactive widget

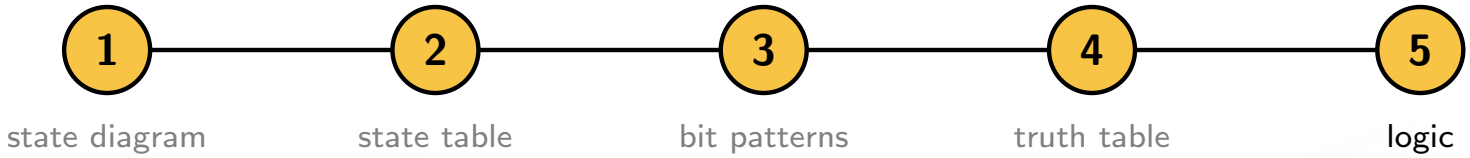
Audience Response

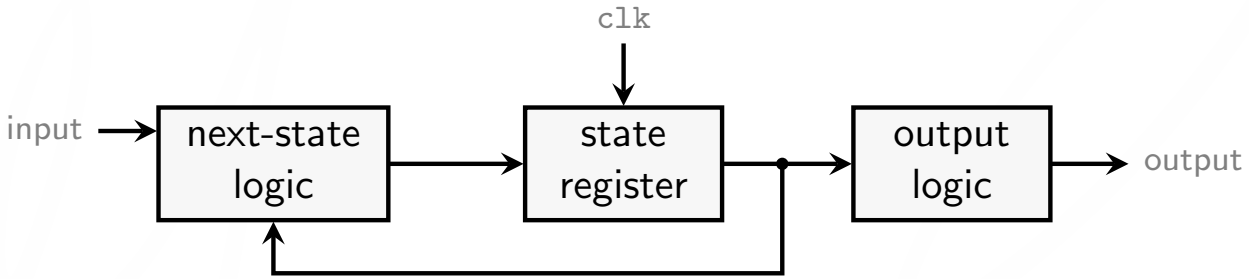
Interactive widget

8

finite state machines

designing an FSM



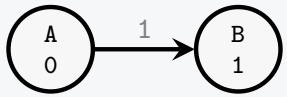


flip-flops needed

$\lceil \log_2 k \rceil$

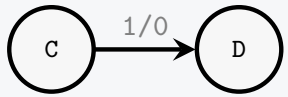
for k states (5 to 8 states need 3)

Moore



output depends on the state only

Mealy

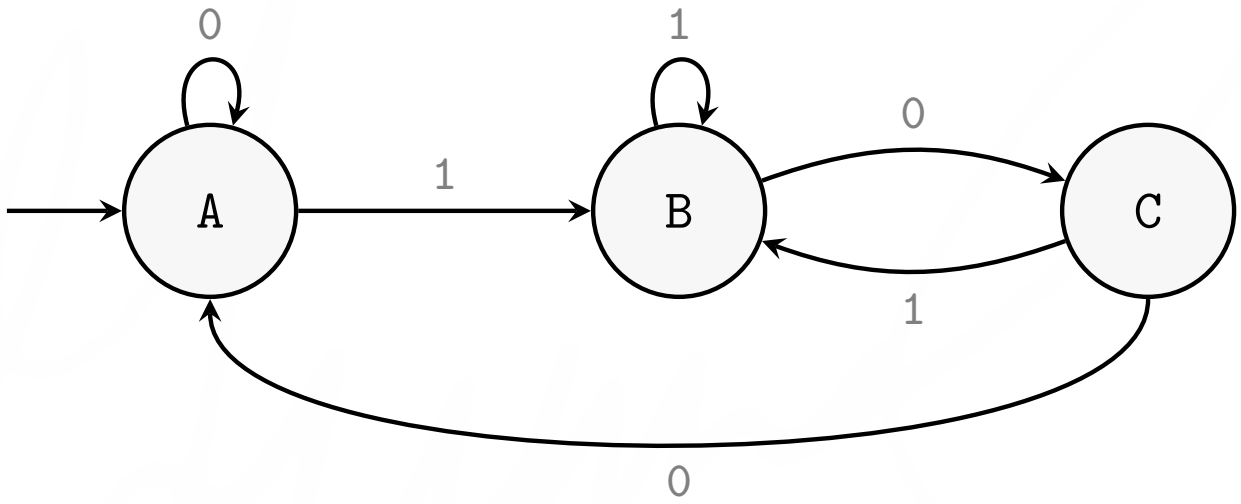


output depends on the state and input

Audience Response

Interactive widget

question 18



the machine starts in A; the input is 1, 0, 1, 1, 0, 1, 0, 0, 1

Audience Response

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question 19

Q_1	Q_0	x	D_1	D_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

a 2-bit counter: it counts up when $x = 1$ and holds when $x = 0$

which equation gives D_1?

Which equation gives $\mathtt{D}_1$?

A $\mathtt{Q}_1 \oplus \mathtt{x}$

B $\mathtt{Q}_0\mathtt{x}$

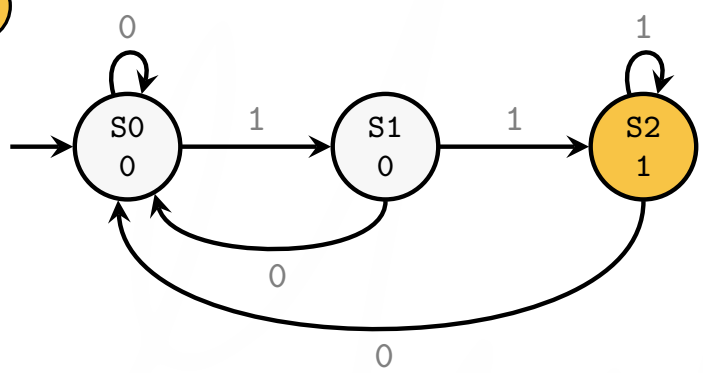
C $\texttt{Q}|\textttt{1} \oplus |\textttt{Q}\texttt{0} \texttt{x}$

D $\texttt{Q}|\textttt{1} + |\textttt{Q}\texttt{0} \texttt{x}$

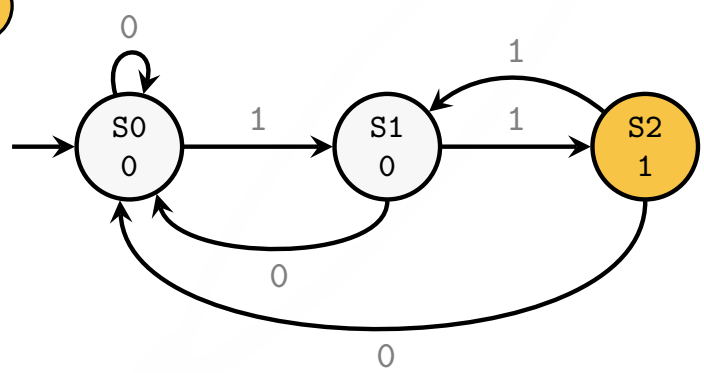
question 20

output 1 exactly when the last two inputs were both 1

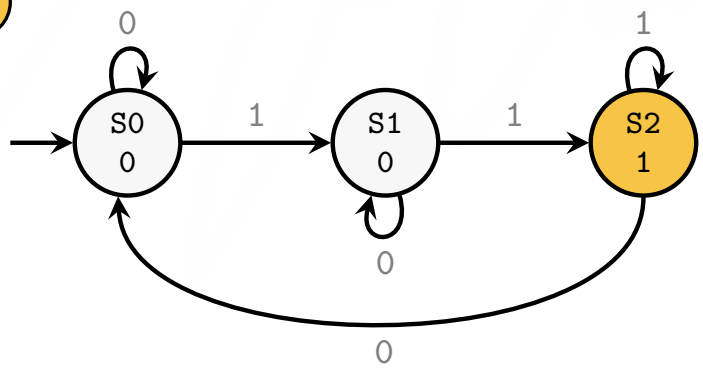
1



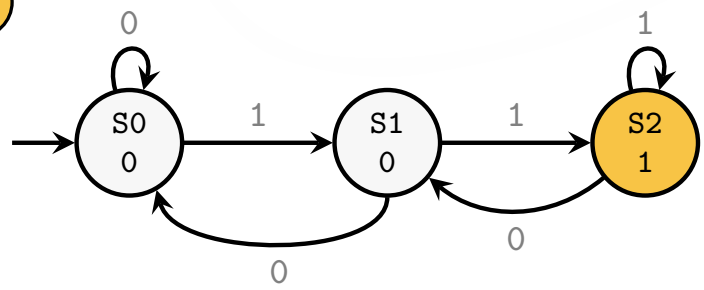
2



3



4



6 responses

Which diagram outputs `1` exactly when the last two inputs were both `1`?

✓ 1	33% 2 votes	B 2	17% 1 vote
C 3	17% 1 vote	D 4	33% 2 votes

putting it all together

A 2-to-4 decoder takes inputs `a` (high bit) and `b`. Its outputs $\texttt{y}|texttt{1}and|texttt{y}\texttt{2}$ feed an `OR` gate. Which single gate is this circuit?

A AND

B XNOR

C XOR

D NAND

On a 4-bit ALU, `a` = `0011` and `b` = `1010` are signed. What does `a − b` give, and does it overflow?

A `1001`, no overflow

B `0111`, no overflow

C `1001`, overflow

D `1101`, overflow

Two rising-edge T flip-flops start at $\texttt{Q}|texttt{1Q}\texttt{0}$ = `00`, with $\texttt{T}|texttt{0} = |texttt{1}and|texttt{T}\texttt{1} = \texttt{Q}|texttt{0}.Whatis|texttt{Q}\texttt{1Q}_\texttt{0}$ after three clock edges?

A `01`

B `10`

C `11`

D `00`

A Moore machine outputs `1` after two `1`s in a row. Its states are S0 = `00`, S1 = `01`, S2 = `10` (bits $\texttt{Q}|texttt{1Q}\texttt{0}).On\texttt{x} = \texttt{1} itmovesS0 \to S1 \to S2andstaysinS2; on\texttt{x} = \texttt{0} itreturnstoS0.UsingDflip - flops, whatis \texttt{D}_\texttt{1}$?

A $\texttt{x}\texttt{Q}_0$

B $\texttt{Q}_1 + \texttt{x}$

C $\texttt{x}(\texttt{Q}|texttt{1} + |texttt{Q}\texttt{0})$

D $\texttt{x}\overline{\texttt{Q}_1}$

Simplify $\overline{\overline{a} + \overline{b}} + \overline{a}b$. What is the result, and its gate-input cost G?

A $a + b$, G = 2

B $ab + b$, G = 4

C b, G = 2

D b, G = 0

A 4-bit subtractor computes $a - b$ on **signed** numbers, and its sign bit is used as the test $a < b$. For which pair does this test give the wrong answer?

A `a = 0011, b = 0101`

B `a = 0111, b = 1000`

C `a = 1110, b = 0010`

D `a = 0101, b = 0101`

Two 4-bit signed numbers, `1101` and `0011`, are multiplied into an 8-bit product. What is it?

A `00100111`

B `11111001`

C `11110111`

D `10001001`

For $\mathtt{f}(\mathtt{a}, \mathtt{b}, \mathtt{c}) = \Pi\mathtt{M}(0, 1, 2, 3, 5, 7)$, what is the minimal product of sums?

A $\mathtt{a} + \mathtt{c}$

B $\mathtt{a}, \overline{\mathtt{c}}$

C $\overline{\mathtt{a}} + \mathtt{c}$

D $\mathtt{a}, (\mathtt{b} + \overline{\mathtt{c}})$

A JK flip-flop with J = K = 1 is clocked at 100 MHz. What is the frequency of its output `Q`?

A 200 MHz

B 100 MHz

C 50 MHz

D 25 MHz

A Mealy machine and a Moore machine each output `1` whenever the last three inputs were `1`, `0`, `1` (overlaps allowed). What are the fewest states each needs (Mealy, Moore)?

A (2, 3)

B (3, 3)

C (3, 4)

D (4, 4)

Good luck on the **midterm**!